

From Handcrafted Features to Transformers

A Comparative Study of Music Genre Classification

Course: CS503 Machine Learning

Members: Hardik Rangi Jagrati Gupta Nishit Soni

Institute: Indian Institute of Technology Ropar



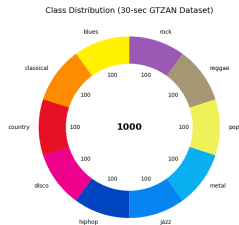
April 2026

Problem Statement and Motivation

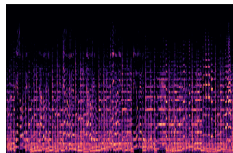
- Music Genre Classification is important for recommendation systems, playlist generation, and music organization.
- Music signals contain overlapping rhythm, harmony, timbre, and melody patterns.
- Classical handcrafted features may miss complex temporal and spectral structures.
- We compare classical ML, deep learning, and transformer-based representations.

Project Goal

To study how feature representation affects music genre classification performance on GTZAN.



GTZAN music genres



Mel-spectrogram representation

Dataset and Project Workflow

Dataset Overview

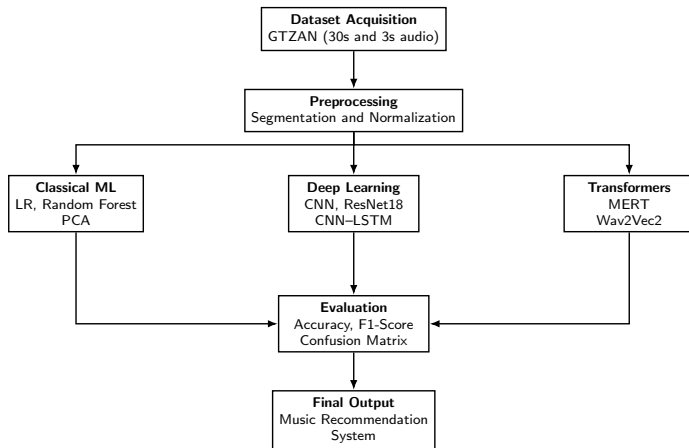
- Dataset: **GTZAN**
- 10 music genres
- 1000 audio clips (30 sec)
- 3-sec version: **9990 samples**
- Split: **70 / 15 / 15**

Dataset Components

- **genres_original/** : raw audio
- **images_original/** : spectrograms
- **features_30_sec.csv** : (1000, 60)
- **features_3_sec.csv** : (9990, 60)

Important

Track-level splitting avoids train-test leakage.



Classical ML, Deep Learning, and Transformer pipelines

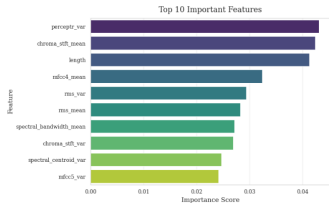
Classical ML and Weighted Hybrid Model

Random Forest + Logistic Regression Weighted Hybrid Fusion

$$P_{final} = \alpha P_{RF} + (1 - \alpha) P_{LR}$$

- **RF:** captures non-linear feature interactions.
- **LR:** Interpretable and efficient probabilistic classifier.
- **Fusion:** combines RF robustness with LR stability.
- **Voting:** track-level aggregation for 3-sec clips.

Feature Importance Using RF



Important contributors: MFCC, chroma, RMS and spectral descriptors.

Model	Dataset	Acc
LR	30-sec	0.710
RF	30-sec	0.690
Hybrid	30-sec	0.745
LR	3-sec	0.760
RF	3-sec	0.775
Hybrid	3-sec	0.810

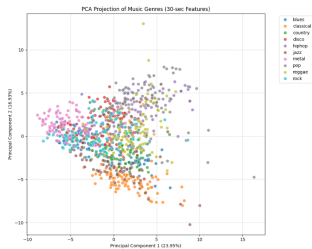
- Hybrid improves over LR and RF on both datasets.
- 3-sec segmentation increases data diversity.
- The hybrid model combines Random Forest's non-linear learning with Logistic Regression's stable generalization using probability-level fusion, improving robustness and prediction reliability while remaining computationally efficient.
- Best classical result achieved using RF-weighted hybrid fusion with $\alpha = 0.6$, giving **81.0%** accuracy on 3-sec audio.

Dimensionality Reduction and Leakage Prevention

PCA: Compact Features

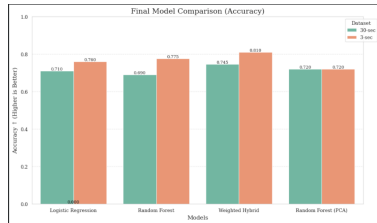
- Redundancy removal
- **95% variance retained**
- Nearly **40% dimension reduction**
- RF was evaluated on reduced features.

Data	Dim.	Acc	F1
30s	57→33	72.0	0.719
3s	57→39	72.0	0.716



PCA projection of music genres

Accuracy Comparison



Leakage Prevention

- Random 3-sec split causes inflated accuracy
- Same parent song may appear in train/test
- Track-level split ensures fair evaluation

PCA Analysis

PCA improved stability for the 30-sec RF model, but reduced 3-sec performance by removing fine spectral details.

Deep Learning with Mel-Spectrograms

Custom CNN

- Input: **Mel-Spectrogram** (128×128)
- 4 Conv blocks: 32, 64, 128, 256 filters
- BatchNorm + ReLU + MaxPooling
- Dense(512) + Dropout(0.4)
- Adam ($lr = 0.0005$, batch=32)
- Accuracy: **68.0%**

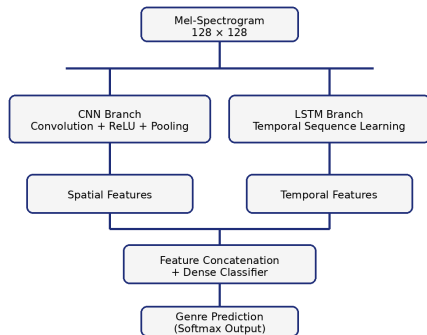
Transfer Learning (ResNet18)

- Pretrained deep CNN architecture
- Extracts richer visual/audio features
- Better generalization than Custom CNN
- Accuracy: **74.83%**

Hybrid Learning

CNN captures **spectral patterns**, while LSTM learns **temporal evolution** in music.

Hybrid CNN-LSTM



Hybrid CNN-LSTM Architecture

- Feature fusion improves robustness
- Captures both spatial and temporal information
- Hybrid CNN-LSTM Accuracy: **78.15%**

Transformers and Music Recommendation System

Transformer Comparison

Model	Pretraining	Acc
Wav2Vec2	Speech	47.68%
MERT	Music	81.33%

Performance Comparison

Model	Type	Acc
LR	Handcrafted	71.0
Hybrid RF+LR	Handcrafted	81.0
Custom CNN	Mel-Spec	68.0
ResNet18	Mel-Spec	74.83
CNN-LSTM	Mel-Spec	78.15
Wav2Vec2	Raw Audio	47.68
MERT	Raw Audio	81.33

Recommendation System

$$\text{sim}(x, y) = \frac{x \cdot y}{||x|| ||y||}$$

- Cosine similarity using **MERT embeddings**
- Captures **tempo, timbre, polyphony**
- Solves **Cold Start Problem**

Comparative Analysis

- **Best Model:** MERT (81.33%)
- **Classical ML:** lightweight, interpretable
- **Deep Learning:** stronger feature learning
- **MERT Embeddings** outperform spectrograms
- **Wav2Vec2:** speech-domain mismatch
- **Limitations:** small GTZAN dataset

Conclusion and Future Directions

Conclusion

- Representation quality is the key factor in genre classification.
- Classical ML gives strong, lightweight, and interpretable baselines.
- CNN-LSTM improves robustness by combining spatial and temporal features.
- **MERT achieved the best accuracy: 81.33%** using music-specific embeddings.

Future Work

- Late fusion of MFCCs and transformer embeddings.
- Temporal attention for identifying important segments.
- Data augmentation using pitch shifting and time stretching.
- Cross-dataset evaluation for improved model generalization.

Final Takeaway

Handcrafted $\xrightarrow{+7.15\%}$ Spectrograms $\xrightarrow{+3.18\%}$ Music Embeddings

Performance improves as representation becomes more expressive and domain-specific.

References

- 1 G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," *IEEE Trans. Speech Audio Process.*, 2002.
- 2 A. Baevski et al., "wav2vec 2.0: A Framework for Self-Supervised Learning," *NeurIPS*, 2020.
- 3 H. Won et al., "MERT: Acoustic Music Understanding Model with Large-Scale Self-Supervised Training," *arXiv preprint arXiv:2306.00107*, 2023.